Training Vectors

$x_1 = \begin{pmatrix}1\\2\\1\end{pmatrix}^T \quad x_2 = \begin{pmatrix}2\\2\\1\end{pmatrix}^T \quad x_3 = \begin{pmatrix}7\\4\\1\end{pmatrix}^T \quad x_4 = \begin{pmatrix}8\\4\\1\end{pmatrix}^T$

$y_1 = 1 \quad y_2 = 1 \quad y_3 = -1 \quad y_4 = -1$

Test Vectors

$T_1 = \begin{pmatrix}2\\3\\1\end{pmatrix}^T \quad T_2 = \begin{pmatrix}4\\5\\1\end{pmatrix}^T \quad T_3 = \begin{pmatrix}6\\2\\1\end{pmatrix}^T$

Use C=1

$$\alpha_1 y_1 + \alpha_2 y_2 + \alpha_3 y_3 + \alpha_4 y_4 = 0$$
$$0 \le \alpha_1 \le C,\ 0 \le \alpha_2 \le C,\ 0 \le \alpha_3 \le C,\ 0 \le \alpha_4 \le C$$

Minimize $\mathcal{L}_D(\alpha) = \frac{1}{2}\sum_{i=1}\sum_{j=1}\alpha_i\alpha_j y_i y_j \left[x_i^T x_j + 1\right]^2 - \sum_{i=1}\alpha_i$

x_1 (illegible)

x_2 (illegible)

x_3 (illegible)

x_4 (illegible)

x_1 x_2 x_3 x_4

(illegible)

x_1 (illegible)

x_2 (illegible)

x_3 (illegible)

x_4 (illegible)

$[x_i \ x_j + 1]$

(illegible)

Minimize : $[+4.5\alpha_1^2 + 12.5\alpha_1\alpha_2 - 72\alpha_1\alpha_3 - 84.5\alpha_1\alpha_4 + 12.5\alpha_2\alpha_1 + 40.5\alpha_2^2 - 264.$

Subject to:

$-[\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4]$

$\sum \alpha_i y_i = 0$

$\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4 = 0$

$\alpha_i \le C$

$\alpha_1 \le 1,\ \alpha_2 \le 1,\ \alpha_3 \le 1,\ \alpha_4 \le 1$

$\alpha_i \ge 0$

$\alpha_1 \ge 0,\ \alpha_2 \ge 0,\ \alpha_3 \ge 0,\ \alpha_4 \ge 0$

Quadratic Equation in α with constraints